Question			Answer	Marks	AOs		Guidance
13	(i)		$f'(x)=\frac{1}{3}\left(27-8x^3\right)^{-2/3}\times(-24x^2)$ $\left[=\frac{-8x^2}{\left(27-8x^3\right)^{2/3}}\right]$ $f'(1.5)=-\frac{8\times1.5^2}{0}$ and dividing by zero zgives the error.	**M1** **A1** **E1** **[3]**	**1.1a** **1.1** **2.4**	Using the chain rule Allow unsimplified Sufficient to say "can't divide by zero" oe	
	(ii)		$\left(27-8x^3\right)^{1/3}=27^{1/3}\left(1-\frac{8}{27}x^3\right)^{1/3}$ $=3\left(1+\left(\frac{1}{3}\right)\left(-\frac{8x^3}{27}\right)+\frac{\left(\frac{1}{3}\right)\left(-\frac{2}{3}\right)}{2!}\left(-\frac{8x^3}{27}\right)^2+\ldots\right)$ $=3-\frac{8x^3}{27}-\frac{64x^6}{2187}+\ldots$	**B1** **M1** **A1** **[3]**	**3.1a** **1.1a** **1.1b**	Dealing with the 27 correctly Using the Binomial expansion substantially correctly Cao	
	(iii)		The binomial expansion is valid for $\left\lvert-8\frac{x^3}{27}\right\rvert<1$ $\lvert x\rvert<1.5$ and the limits of the integral are completely in this interval.	**B1** **E1** **[2]**	**2.4** **2.3**	Allow unsimplified but must use correct modulus notation or equivalent Must indicate that the limits of the integral lie in their interval for which the expansion is valid.	
	(iv)		$\frac{0.25}{2}\left(3+2.6684+2\left(2.9954+2.9625+2.8694\right)\right)$ $=\frac{0.25}{2}\times23.3224=2.9153$	**B1** **M1** **A1** **[3]**	**1.1a** **1.1b** **1.1b**	$h=0.25$ used For sum in the bracket – condone one slip. Allow for 2.92 or better	Values from candidates own calculators may differ in the last decimal place.
	(v)		There is area between the curve and the top of the trapezia, so some area is missing from the estimate.	**E1** **[1]**	**2.4**	Allow for any sensible explanation eg the trapezia are under the curve	"The curve is concave downwards" on its own is not quite enough